MATH317 · NUMERICAL METHODS

Midterm 461 Practice

2024 · 60 minutes · 25 marks

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01 Questions

Answer each question in the space below it. The full solution is in a separate file.

Q1 6 + 5 M

1. A program approximates square roots:

x = 4: true √x = 2.0000, computed 2.1000.

x = 10: true √x = 3.1623, computed 3.1600.

a) Define the absolute and relative errors. b) Compute the absolute errors. c) Compute the relative errors as percentages.

2. Given $f(x) = x^3 - 2x - 1$ and $x_0 = 1.5$, apply Newton's method for two iterations.

Q2 6 M

Determine the single-precision machine representation of the decimal number 1259.125.

Q3 2 + 6 M

Given $f(x) = e^x - 4x^2$.

a) Prove that f(x) has exactly one real root in the interval [0, 1].

b) Find a root on [0, 1] by using the bisection method. Terminate the iterations if they exceed nmax = 2 or if the error bound is less than 0.5×10^{-6}.